
#Seaborn Task 1

```
import seaborn as sns
```

```
tips = sns.load_dataset("tips")
print(tips.head())
print(tips.info())
```

```
===== RESTART: C:/Users/krsb/OneDrive/Desktop/internsh.
```

```
   total_bill  tip  sex smoker  day  time  size
0      16.99  1.01 Female    No  Sun  Dinner    2
1      10.34  1.66  Male    No  Sun  Dinner    3
2      21.01  3.50  Male    No  Sun  Dinner    3
3      23.68  3.31  Male    No  Sun  Dinner    2
4      24.59  3.61 Female    No  Sun  Dinner    4
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 244 entries, 0 to 243
```

```
Data columns (total 7 columns):
```

#	Column	Non-Null Count	Dtype
0	total_bill	244 non-null	float64
1	tip	244 non-null	float64
2	sex	244 non-null	category
3	smoker	244 non-null	category
4	day	244 non-null	category
5	time	244 non-null	category
6	size	244 non-null	int64

```
dtypes: category(4), float64(2), int64(1)
```

```
memory usage: 7.4 KB
```

```
None
```

#Seaborn Task 2

```
import seaborn as sns
```

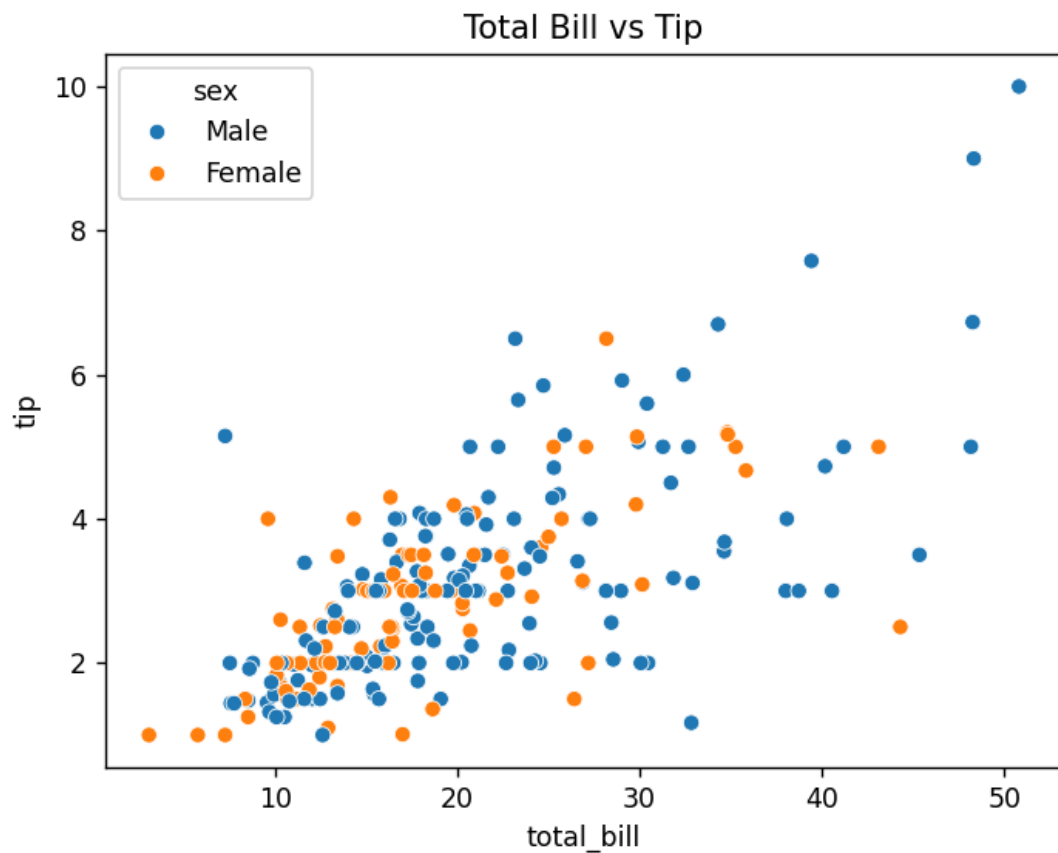
```
import matplotlib.pyplot as plt
```

```
tips = sns.load_dataset("tips")
```

```
sns.scatterplot(data=tips, x="total_bill", y="tip", hue="sex")
```

```
plt.title("Total Bill vs Tip")
```

```
plt.show()
```

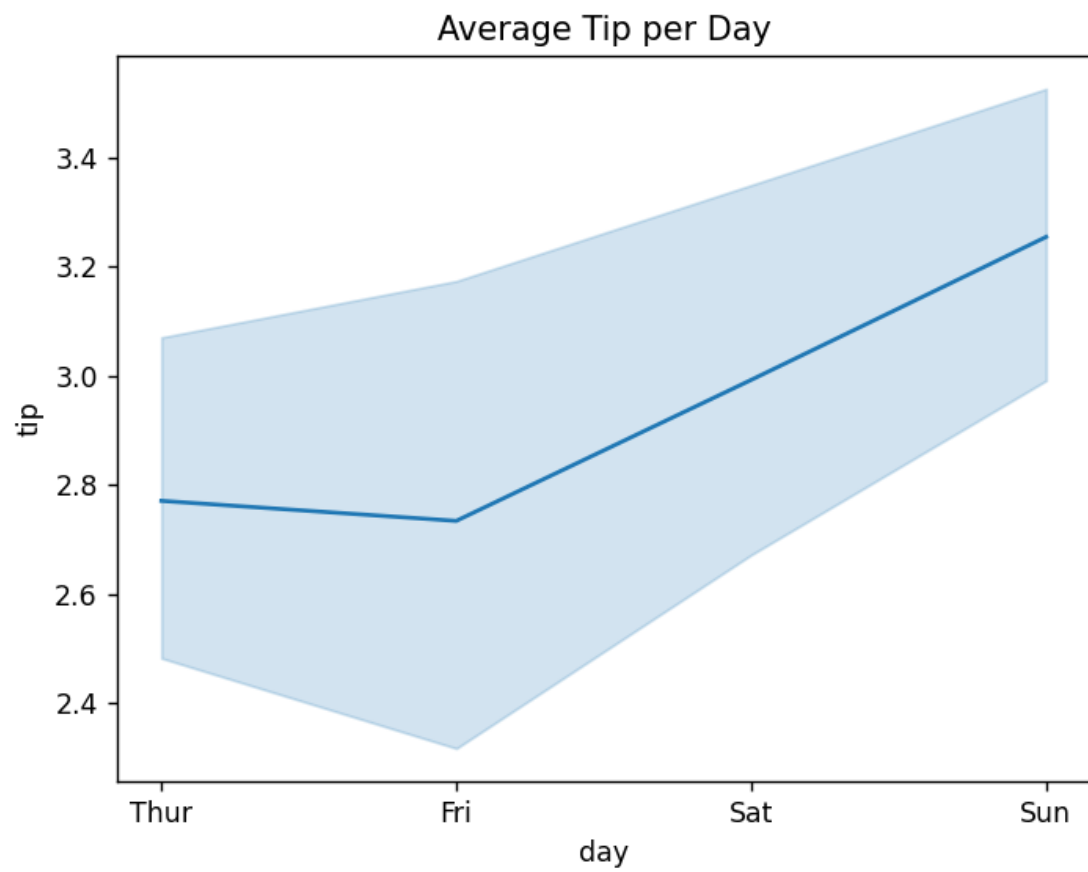


#Seaborn Task 2b

```
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")

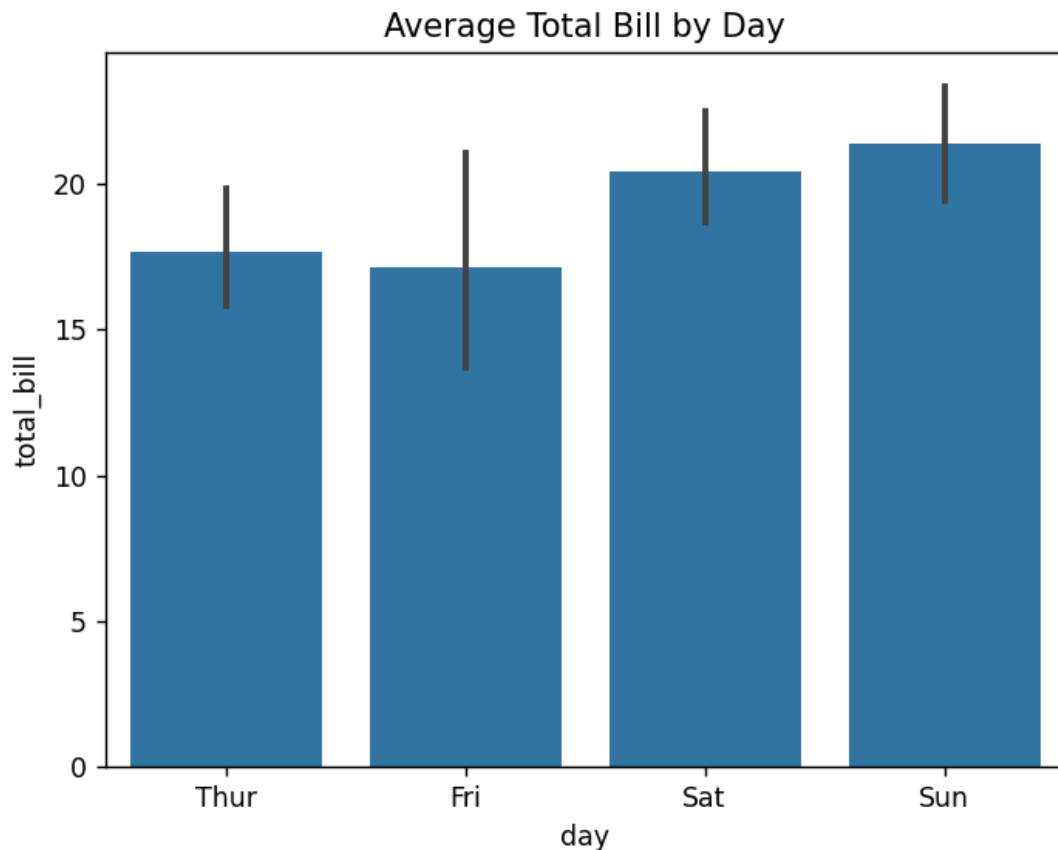
sns.lineplot(data=tips, x="day", y="tip", estimator="mean")
plt.title("Average Tip per Day")
plt.show()
```



```
#Seaborn Task 2c
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")

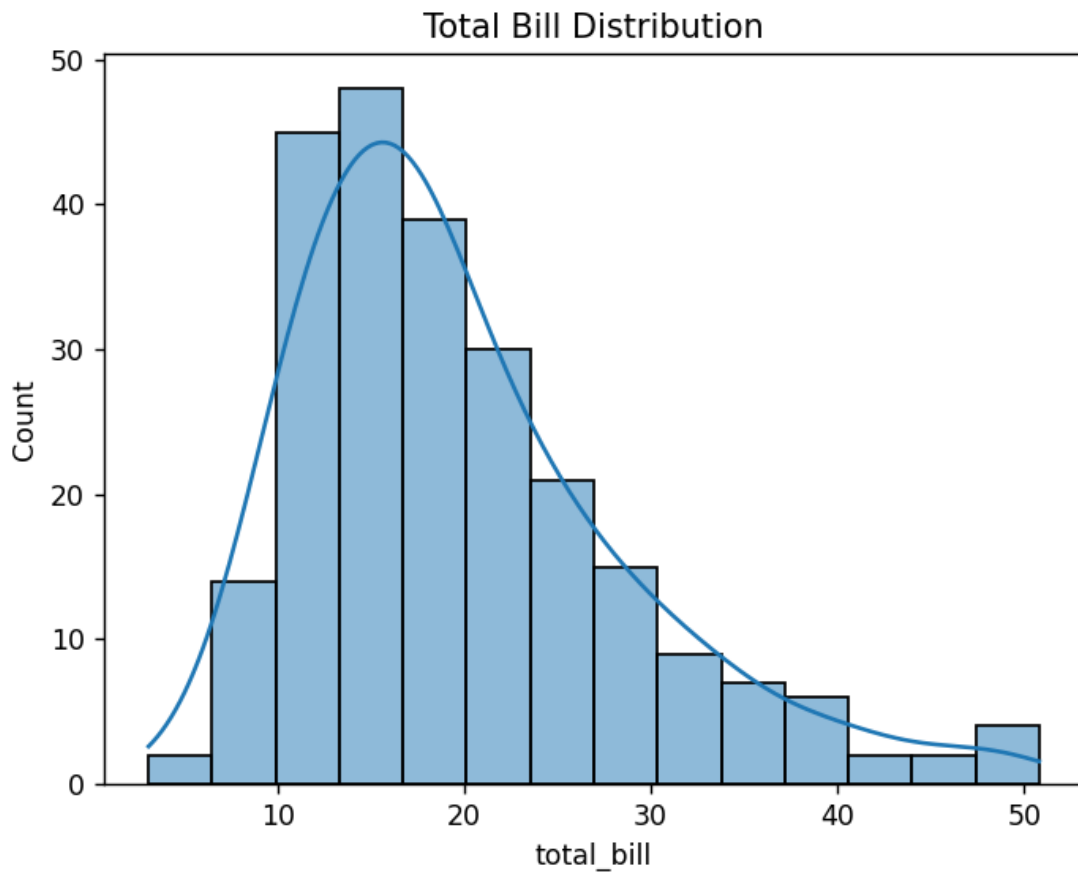
sns.barplot(data=tips, x="day", y="total_bill")
plt.title("Average Total Bill by Day")
plt.show()
```



```
#Seaborn Task 2d
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")

sns.histplot(tips["total_bill"], kde=True)
plt.title("Total Bill Distribution")
plt.show()
```



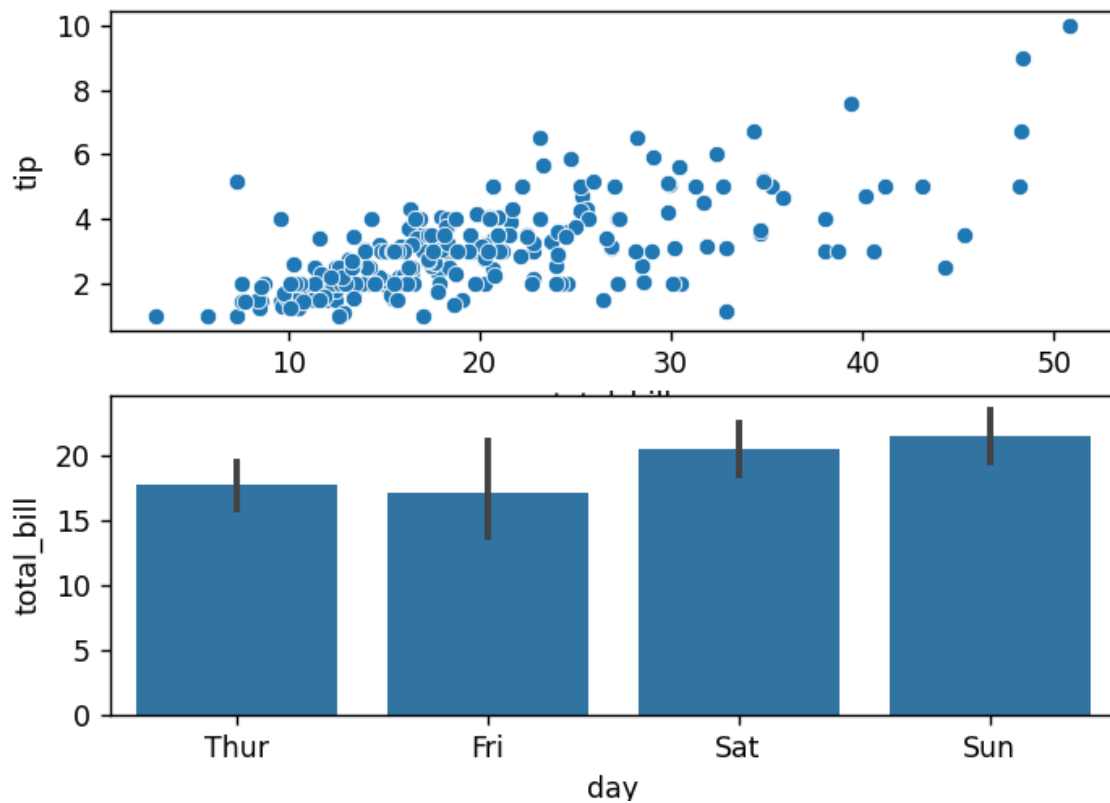
```
#Seaborn Task 3
|
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")

fig, axes = plt.subplots(2, 1)

sns.scatterplot(data=tips, x="total_bill", y="tip", ax=axes[0])
sns.barplot(data=tips, x="day", y="total_bill", ax=axes[1])

plt.show()
```

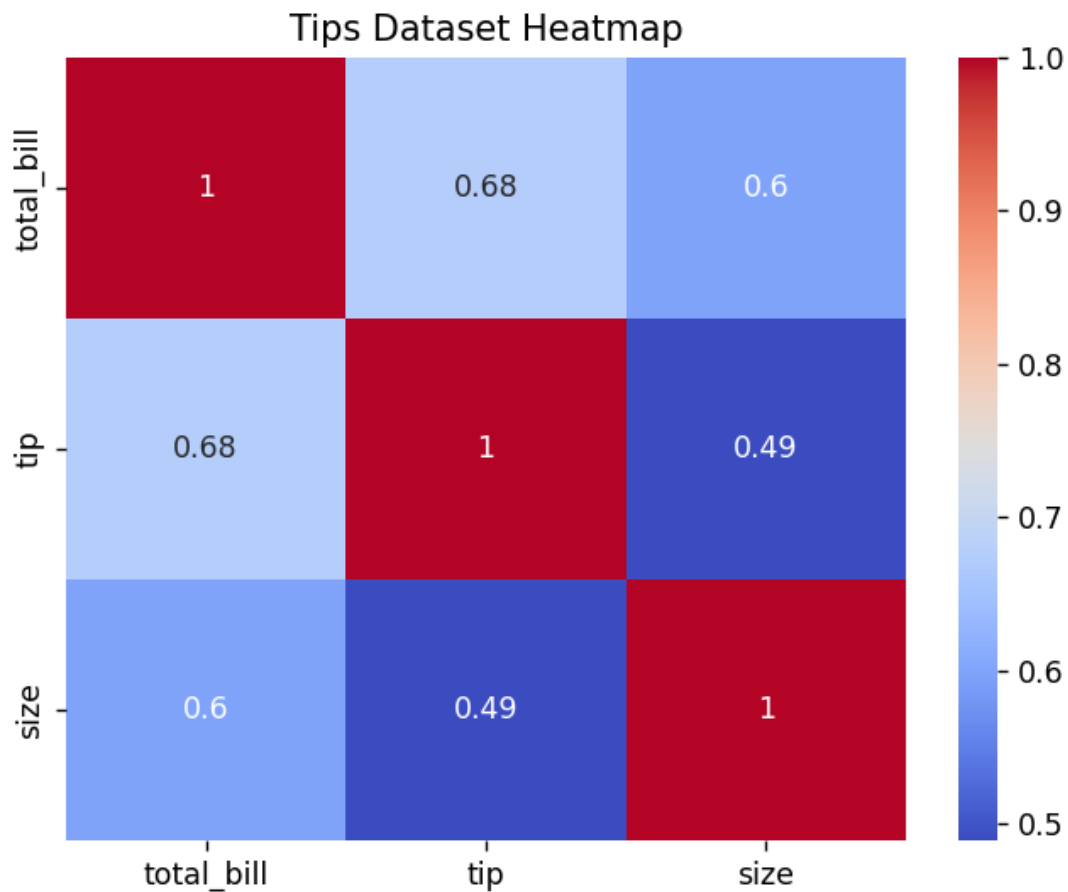


#Seaborn Task 4

```
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")

corr = tips.corr(numeric_only=True)
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Tips Dataset Heatmap")
plt.show()
```



#Seaborn Task 5

```
import seaborn as sns
import matplotlib.pyplot as plt

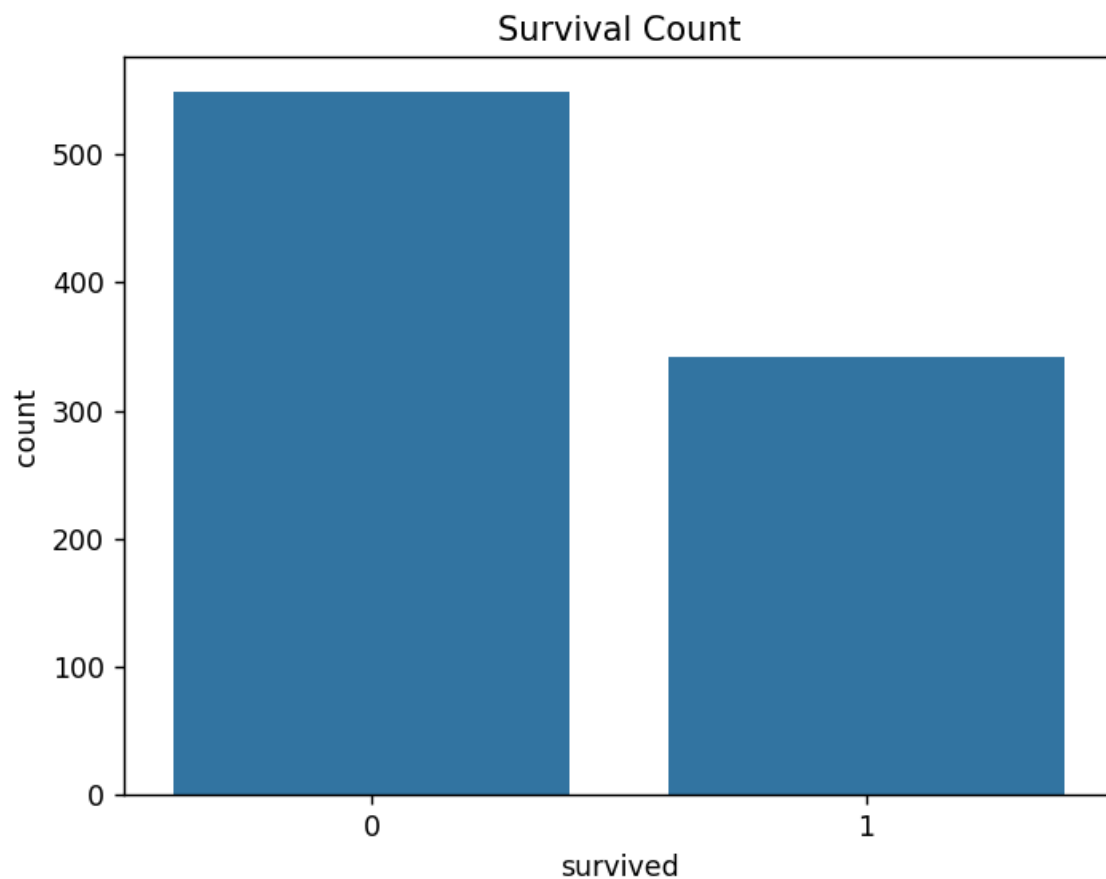
titanic = sns.load_dataset("titanic")

sns.countplot(data=titanic, x="survived")
plt.title("Survival Count")
plt.show()

sns.histplot(titanic["age"], kde=True)
plt.title("Age Distribution")
plt.show()

sns.countplot(data=titanic, x="sex", hue="survived")
plt.title("Survival by Gender")
plt.show()

sns.heatmap(titanic.corr(numeric_only=True), annot=True, cmap="viridis")
plt.title("Titanic Heatmap")
plt.show()
```



```
#Pillow Task 1&2

from PIL import Image

img = Image.open("car.jpg")
img.show()

print(img.format, img.size, img.mode)
```



```
===== RESTART: C:/Users/krsb/OneDriv
JPEG (3840, 2160) RGB
```



```
#Pillow Task 3
```

```
from PIL import Image
```

```
img = Image.open("car.jpg")  
resized = img.resize((300, 300))  
resized.show()
```



```
#Pillow Task 4
```

```
from PIL import Image
```

```
img = Image.open("car.jpg")  
gray = img.convert("L")  
gray.show()
```



#Pillow Task 5

```
from PIL import Image
```

```
img = Image.open("car.jpg")
```

```
bw = img.convert("1")
```

```
bw.show()
```



```
#Pillow Task 6
```

```
from PIL import Image
```

```
img = Image.open("car.jpg")  
rotated = img.rotate(90)  
rotated.show()
```



```
#Pillow Task 7
```

```
from PIL import Image
```

```
img = Image.open("car.jpg")  
  
cropped = img.crop((0, 0, img.width//2, img.height//2))  
cropped.show()
```



```
#Pillow Task 8
```

```
from PIL import Image
import os
```

```
print("Current folder:", os.getcwd())
```

```
img = Image.open("car.jpg")
img.save("saved_image.jpg")
```

```
print("Image saved successfully!")
```

```
===== RESTART: C:/Users/krsb/OneDrive/Desktop/internship/DAY 22 TASK.}
Current folder: C:\Users\krsb\OneDrive\Desktop\internship
Image saved successfully!
|
```

```
#Pillow Task 9
```

```
from PIL import Image
```

```
img = Image.open("car.jpg")
```

```
img.thumbnail((100, 100))
```

```
img.show()
```



```
#Pillow Task 10
```

```
from PIL import Image, ImageFilter
```

```
img = Image.open("car.jpg")
```

```
blur = img.filter(ImageFilter.GaussianBlur(radius=12))
```

```
blur.show()
```



```
#Pillow Task 11
```

```
from PIL import Image, ImageFilter
```

```
img = Image.open("car.jpg")
```

```
sharp = img.filter(ImageFilter.SHARPEN)
```

```
sharp.show()
```



#Pillow Task 12

```
from PIL import Image, ImageFilter
```

```
img = Image.open("car.jpg")
```

```
edge = img.filter(ImageFilter.FIND_EDGES)
```

```
edge.show()
```



```
#Pillow Task 13
```

```
from PIL import Image  
import os
```

```
print("Saving image in:", os.getcwd())
```

```
img = Image.open("car.jpg")  
img = img.resize((200, 200))  
img = img.convert("L")  
img = img.rotate(90)  
img.save("final_image.jpg")
```

```
print("final_image.jpg saved successfully")  
img.show()
```

```
===== RESTART: C:/Users/krsb/OneDrive/Desktop/internship/DAY 22 TASK.py =====  
Saving image in: C:\Users\krsb\OneDrive\Desktop\internship  
final_image.jpg saved successfully
```

